Calculus 2

Section 2

9. $\int (3x-2)^{10}\,dx$

Let $u = 3x - 2$, $du = 3dx \implies dx = \frac{du}{3}$

$$\int u^{10} \cdot \frac{du}{3} = \frac{1}{3} \cdot \frac{u^{11}}{11} + C$$

$$= \frac{1}{33}(3x-2)^{11} + C$$

$$\boxed{\frac{1}{33}(3x-2)^{11} + C}$$

10. $\int \frac{dx}{5-2x}$

Let $u = 5 - 2x$, $du = -2dx \implies dx = -\frac{du}{2}$

$$\int \frac{1}{u} \cdot \left(-\frac{du}{2}\right) = -\frac{1}{2}\ln|u| + C$$

$$= -\frac{1}{2}\ln|5-2x| + C$$

$$\boxed{-\frac{1}{2}\ln|5-2x| + C}$$

11. $\int \frac{x\,dx}{\sqrt{20x^2+1}}$

Let $u = 20x^2 + 1$, $du = 40xdx \implies xdx = \frac{du}{40}$

$$\int \frac{1}{\sqrt{u}} \cdot \frac{du}{40} = \frac{1}{40} \int u^{-1/2} du$$

$$= \frac{1}{40} \cdot 2\sqrt{u} + C = \frac{\sqrt{20x^2+1}}{20} + C$$

$$\boxed{\frac{\sqrt{20x^2+1}}{20} + C}$$

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12. $\int \cos^4(9x+6)\sin(9x+6)\,dx$

Let $u = \cos(9x+6)$, $du = -9\sin(9x+6)dx \implies \sin(9x+6)dx = -\frac{du}{9}$

$$\int u^4 \cdot \left(-\frac{du}{9}\right) = -\frac{1}{9} \cdot \frac{u^5}{5} + C$$

$$= -\frac{1}{45}\cos^5(9x+6) + C$$

$$\boxed{-\frac{1}{45}\cos^5(9x+6) + C}$$

13. $\int \sec^2(7x+3)\tan^2(7x+3)\,dx$

Let $u = \tan(7x+3)$, $du = 7\sec^2(7x+3)dx \implies \sec^2(7x+3)dx = \frac{du}{7}$

$$\int u^2 \cdot \frac{du}{7} = \frac{1}{7} \cdot \frac{u^3}{3} + C$$

$$= \frac{1}{21}\tan^3(7x+3) + C$$

$$\boxed{\frac{1}{21}\tan^3(7x+3) + C}$$

14. $\int \frac{\cos x - \sin x}{\cos x + \sin x} dx$

Notice $d(\cos x + \sin x) = (-\sin x + \cos x)dx$ — exactly the numerator.
So $\frac{\cos x - \sin x}{\cos x + \sin x} dx = \frac{du}{u}$ with $u = \cos x + \sin x$:

$$\int \frac{du}{u} = \ln|u| + C$$

$$\boxed{\ln|\cos x + \sin x| + C}$$

15. $\int \frac{1}{x} \ln(x^2)\, dx$

$\ln(x^2) = 2\ln x$

$$\int \frac{1}{x} \cdot 2\ln x\, dx = 2\int \frac{\ln x}{x} dx$$

Let $u = \ln x$, $du = \frac{dx}{x}$

$$2\int u\, du = 2 \cdot \frac{u^2}{2} + C = (\ln x)^2 + C$$

$$\boxed{(\ln x)^2 + C}$$

16. $\int \tan x \ln|\cos x|\, dx$

Let $u = \ln|\cos x|$, $du = \frac{-\sin x}{\cos x} dx = -\tan x dx$
So $\tan x dx = -du$

$$\int u \cdot (-du) = -\frac{u^2}{2} + C$$

$$= -\frac{1}{2}(\ln|\cos x|)^2 + C$$

$$\boxed{-\frac{1}{2}(\ln|\cos x|)^2 + C}$$

17. $\int \frac{\tan^3 x}{\cos^2 x}\, dx$

$\frac{1}{\cos^2 x} = \sec^2 x$, so integrand is $\tan^3 x \sec^2 x$.
Let $u = \tan x$, $du = \sec^2 x dx$

$$\int u^3\, du = \frac{u^4}{4} + C = \frac{\tan^4 x}{4} + C$$

$$\boxed{\frac{\tan^4 x}{4} + C}$$

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18. $\int \frac{\sqrt{1+\ln x}}{x}\, dx$

Let $u = 1 + \ln x$, $du = \frac{dx}{x}$

$$\int u^{1/2}\, du = \frac{u^{3/2}}{3/2} + C = \frac{2}{3}(1 + \ln x)^{3/2} + C$$

$$\boxed{\frac{2}{3}(1 + \ln x)^{3/2} + C}$$

19. $\int \frac{\sqrt{8+\tan x}}{1-\sin^2 x} dx$

$1 - \sin^2 x = \cos^2 x$, and $\frac{1}{\cos^2 x} = \sec^2 x$.
So integrand: $\sqrt{8 + \tan x} \cdot \sec^2 x$.
Let $u = 8 + \tan x$, $du = \sec^2 x dx$

$$\int u^{1/2} du = \frac{2}{3} u^{3/2} + C = \frac{2}{3}(8 + \tan x)^{3/2} + C$$

$$\boxed{\frac{2}{3}(8 + \tan x)^{3/2} + C}$$

20. $\int \frac{\tan^{-1} x}{1+x^2} dx$

Let $u = \tan^{-1} x$, $du = \frac{dx}{1+x^2}$

$$\int u\, du = \frac{u^2}{2} + C$$

$$\boxed{\frac{(\tan^{-1} x)^2}{2} + C}$$

21. $\int \frac{\sin 2x}{\sqrt{1+\sin^2 x}} dx$

Note: $\sin 2x = 2 \sin x \cos x$.

Let $u = \sin^2 x$, $du = 2 \sin x \cos x dx = \sin 2x dx$.

$$\int \frac{du}{\sqrt{1+u}} = \int (1+u)^{-1/2} du = 2\sqrt{1+u} + C = 2\sqrt{1+\sin^2 x} + C$$

Answer:

$$\boxed{2\sqrt{1+\sin^2 x} + C}$$

22. $\int \frac{\sin 2x}{\sin^2 x} dx$

$\sin 2x = 2 \sin x \cos x$

$$\frac{\sin 2x}{\sin^2 x} = \frac{2 \sin x \cos x}{\sin^2 x} = 2 \cot x$$

$$\int 2 \cot x\, dx = 2 \ln |\sin x| + C$$

$$\boxed{2 \ln |\sin x| + C}$$